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Demography predicts genetic effective size in a desert stream fish community

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Life history traits, demographic analyses, and summary of microsatellite loci

Table S.1. Life history traits and detection probabilities (*p*) for Gila River fishes

Species	AL	α	L_{mat} (mm)	L_{max} (mm)	Fecundity	Parental care	mean *p*	*p* range
Agosia chrysogaster	2	1	55	100	3.0	2	0.96	0.86 - 1.00
Gila robusta	10	3	100	405	4.4	1	0.80	0.67 - 1.00
Meda fulgida	1	1	52	91	3.1	1	1.00	1.00-1.00
Rhinichthys cobitis	1	1	45	65	2.8	3	0.92	0.86 - 1.00
Rhinichthys osculus	1	2	66	110	2.7	3	0.91	0.80 - 1.00
Catostomus insignis	11	4	300	800	4.1	1	0.99	0.92 - 1.00
Pantosteus clarkii	13	2	120	330	4.0	1	1.00	1.00 - 1.00
Ameiurus natalis	4	3	233	470	3.6	4	0.87	0.50 - 1.00
Micropterus dolomieu	10.5	3.5	205	690	4.1	4	0.85	0.50 - 1.00

Note – Life history traits include adult lifespan (AL, [maximum age less age at maturity]), age at maturity (α), length at maturity (L_{mat}), maximum reported length (L_{max}), fecundity and parental care as compiled by Olden et al. (2006). Fecundity total number of eggs or offspring per breeding season expressed as $\log_{10}$. Parental care metric describes total energetic contribution of parents to offspring ranging from least (1) to most (4). Detection probabilities (*p*) were estimated across 6 of the 16 sample locations for each of n = 7 multi-pass sampling events (see Whitney et al. [2016]).

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Table S.2. Microsatellite loci summary information for Gila River fishes.

						Hardy-Weinberg Expectations				Linkage Disequilibrium	
Species	n (N)	# Loci	He	Ho	A range	# at P=0.05	Pcor	# at Pcor	%	# sig	Total # tests
A. chrysogaster	397 (13)	10	0.759	0.746	7 - 36	14	0.000385	4	0.031	2	574
G. robusta	80 (5)	10	0.627	0.616	3 - 20	5	0.001	1	0.02	2	169
M. fulgida	170 (6)	10	0.804	0.734	10 - 28	13	0.000833	9	0.15	3	243
R. cobitis	112 (4)	10	0.73	0.677	2 - 56	12	0.00125	6	0.15	1	171
R. osculus	277 (8)	10	0.705	0.661	4 - 39	15	0.000625	3	0.038	2	342
C. insignis	292 (12)	9	0.816	0.825	8 - 28	7	0.000463	0	0	4	324
P. clarkii	344 (12)	10	0.77	0.758	9 - 29	16	0.000417	1	0.008	10	531
A. natalis	122 (6)	9	0.309	0.333	2 - 7	0	0.000926	0	0	3	126
M. dolomieu	195 (10)	9	0.469	0.45	2 - 5	2	0.000556	0	0	2	216

Note – Sample size in total individuals (n) and number of sample locations (N) for microsatellite genotype data previously published in Pilger et al. (2017). Included is the number of loci screened per species, mean expected and observed heterozygosity, range in number of alleles and numbers of significant tests for Hardy-Weinberg proportions and linkage disequilibrium. Individual genotypes are available at https://doi.org/10.5061/dryad.d5j4m.

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Table S.3. Local extirpation and detection parameter estimates for Gila River fishes using program PRESENCE (Hines 2006).

Species	n	Model	AIC[2]	ε	SE	p	SE
A. chrysogaster	5	ψ(.)γ(.)ε(.)*p*(.)[1]	87.34	0.04	0.026	0.93	0.032
		ψ(.)γ(.)ε(.)*p*(fix)	88.20	0.06	0.033	0.963	fixed[3]
G. robusta	4	ψ(.)γ(.)ε(.)*p*(.)[1]	86.39	0.28	0.066	1.00	0.000
		ψ(.)γ(.)ε(.)*p*(fix)[1]	90.58	0.00	-	0.80	fixed[3]
M. fulgida	5	ψ(.)γ(.)ε(.)*p*(.)[1]	115.01	0.07	0.047	0.86	0.066
		ψ(.)γ(.)ε(.)*p*(fix)[1]	119.04	0.21	0.052	1.00	fixed[3]
R. cobitis	5	ψ(.)γ(.)ε(.)*p*(.)[1]	99.25	0.05	0.028	0.89	0.047
		ψ(.)γ(.)ε(.)*p*(fix)	101.36	0.06	0.033	0.92	fixed[3]
R. osculus	3	ψ(.)γ(.)ε(.)*p*(.)	70.24	0.09	0.056	0.97	0.053
		ψ(.)γ(.)ε(.)*p*(fix)	70.89	0.07	0.047	0.91	fixed[3]
C. insignis	5	ψ(.)γ(.)ε(.)*p*(.)[1]	23.68	0.01	.010	1.00	0.000
		ψ(.)γ(.)ε(.)*p*(fix)	46.79	0.00	0.00	0.99	fixed[3]
P. clarkii	5	ψ(.)γ(.)ε(.)*p*(.)[1]	58.01	0.07	0.026	1.00	0.000
		ψ(.)γ(.)ε(.)*p*(fix)	58.01	0.07	0.026	1.00	fixed[3]
A. natalis	4	ψ(.)γ(.)ε(.)*p*(.)	91.07	0.11	0.079	0.90	0.108
		ψ(.)γ(.)ε(.)*p*(fix)	91.17	0.10	0.054	0.87	fixed[3]
M. dolomieu	4	ψ(.)γ(.)ε(.)*p*(.)	95.61	0.15	0.050	1.00	0.000
		ψ(.)γ(.)ε(.)*p*(fix)	103.68	0.12	0.047	0.85	fixed[3]

Note – n is the number of detection histories assessed. Model parameters are proportion of sites occupied (ψ), colonization probability (γ), extirpation probability (ε) and detection probability (*p*). [1] Model may not have achieved convergence during numerical optimization. [2] Model AIC values are not corrected for small sample size. [3] Detection probabilities were fixed to those reported in Table S.1. Bolded estimates of ε were used for downstream analyses.

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Estimating migration rates using BIMr

Contemporary gene flow (*m*, migration rate) was estimated for each species according to the assignment model implemented in BIMr 1.0 (Faubet and Gaggiotti 2003) using Bayesian inference. Because of the generally low level of differentiation observed for Gila River fishes and since the reliability of *m* estimates decreases if $F_{ST} < 0.01$, samples were combined across sample locations if pairwise $F_{ST} \leq 0.01$ between locations. Pairwise F_{ST} values for each species are provided in supplementary material of Pilger et al. (2017). For each species, 10 independent MCMC runs were initiated with a burnin of 10^6 iterations and 20,000 samples were taken from the chain at a thinning interval of 100 (total post-burnin iterations = $2x10^6$). Twenty pilot runs of 1,000 iterations were used to set parameter proposal values and program default values were used for parameter priors. We used Bayesian deviance as a measure of model fit and used *m* estimates from the run that exhibited lowest deviance for the assignment component of total deviance $D_{assign} = -2\log\Pr(M \mid S; m)$ (Faubet and Gaggiotti 2003).

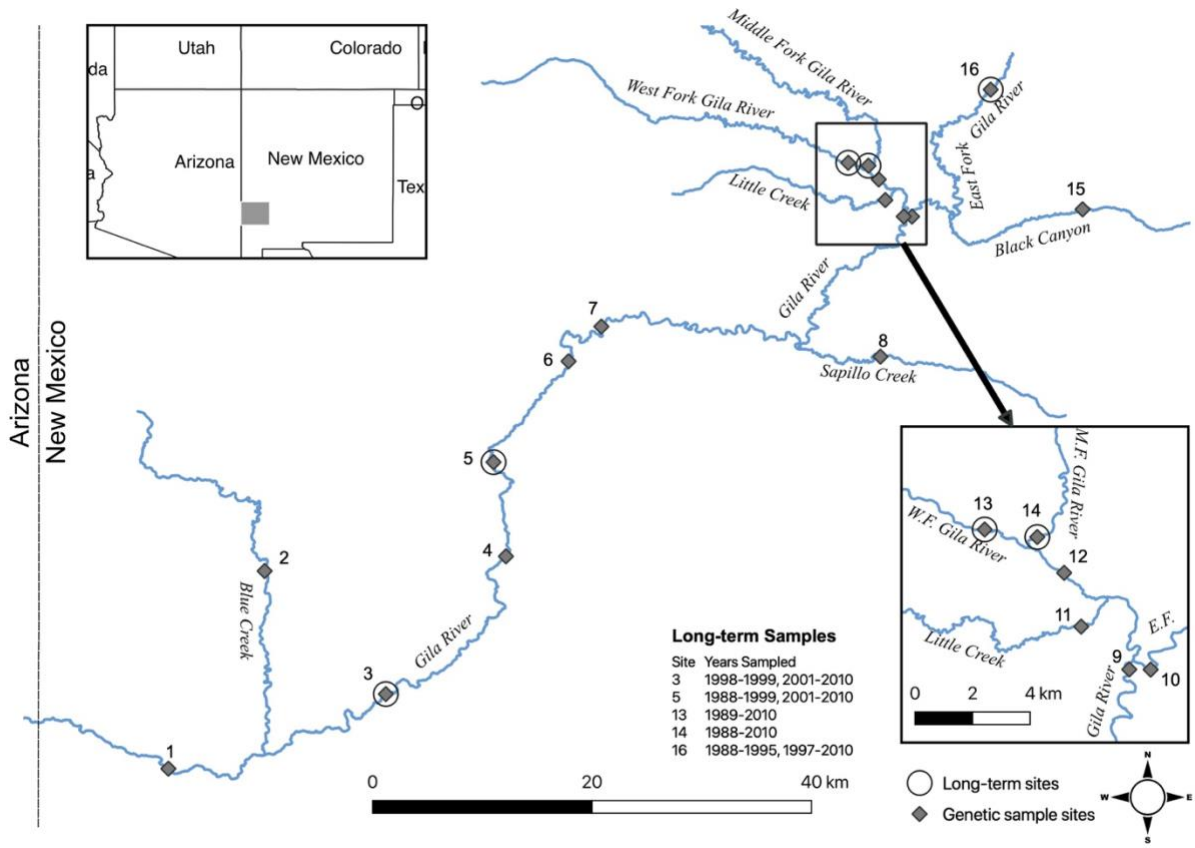

Figure S.1. Locations in the upper Gila River catchment where demographic and genetic data were collected on nine fish species to evaluate metapopulation effective size models.

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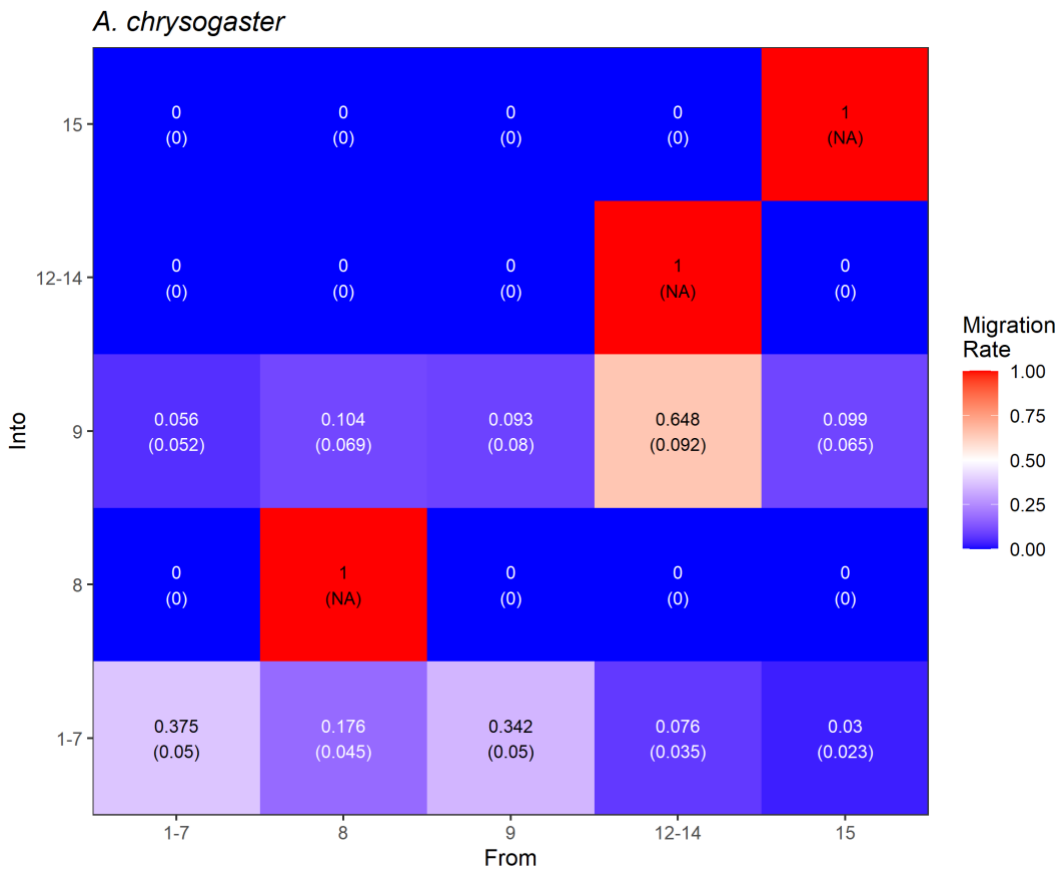

Figure S.2. Migration matrix for *A. chrysogaster*. Posterior means (standard errors), estimated using Bayesian assignment, represent the fraction of individuals at locations along the y-axis that are migrants derived from locations along the x-axis per generation. See Figure S.1 for sample locations corresponding the labels along the x- and y-axes.

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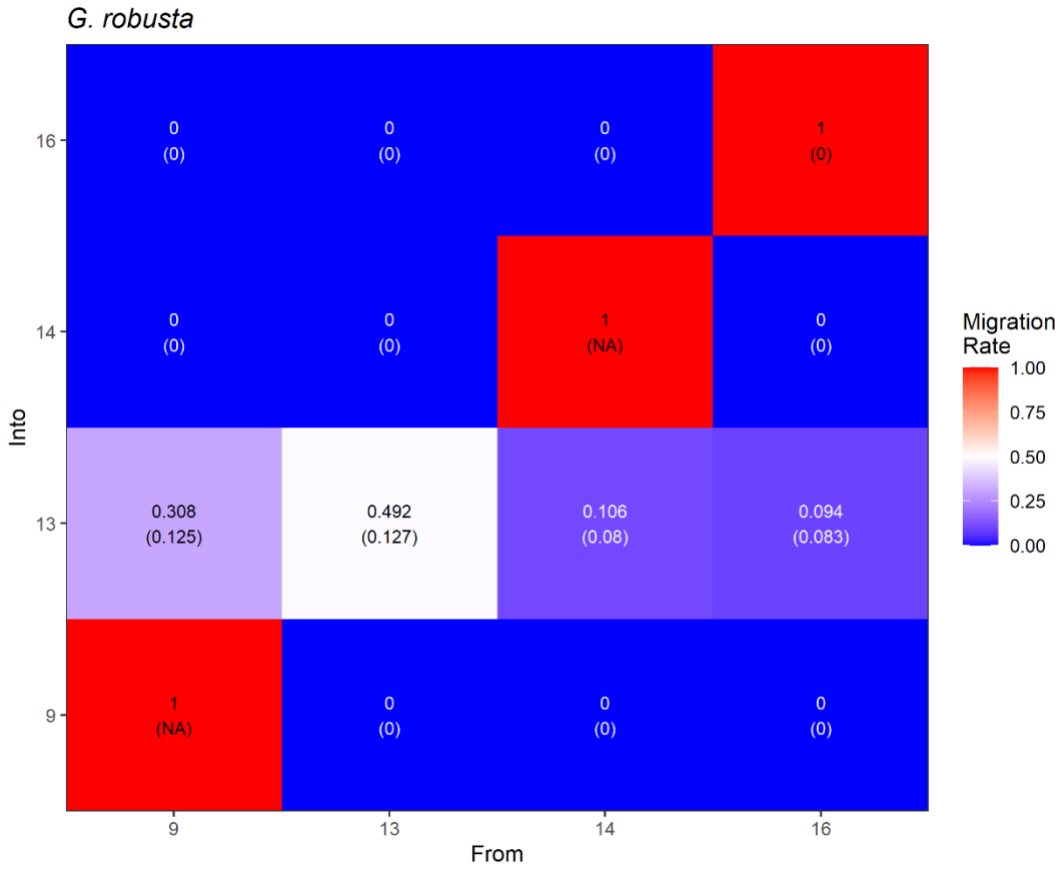

Figure S.3. Migration matrix for *G. robusta*. Posterior means (standard errors), estimated using Bayesian assignment, represent the fraction of individuals at locations along the y-axis that are migrants derived from locations along the x-axis per generation. See Figure S.1 for sample locations corresponding the labels along the x- and y-axes.

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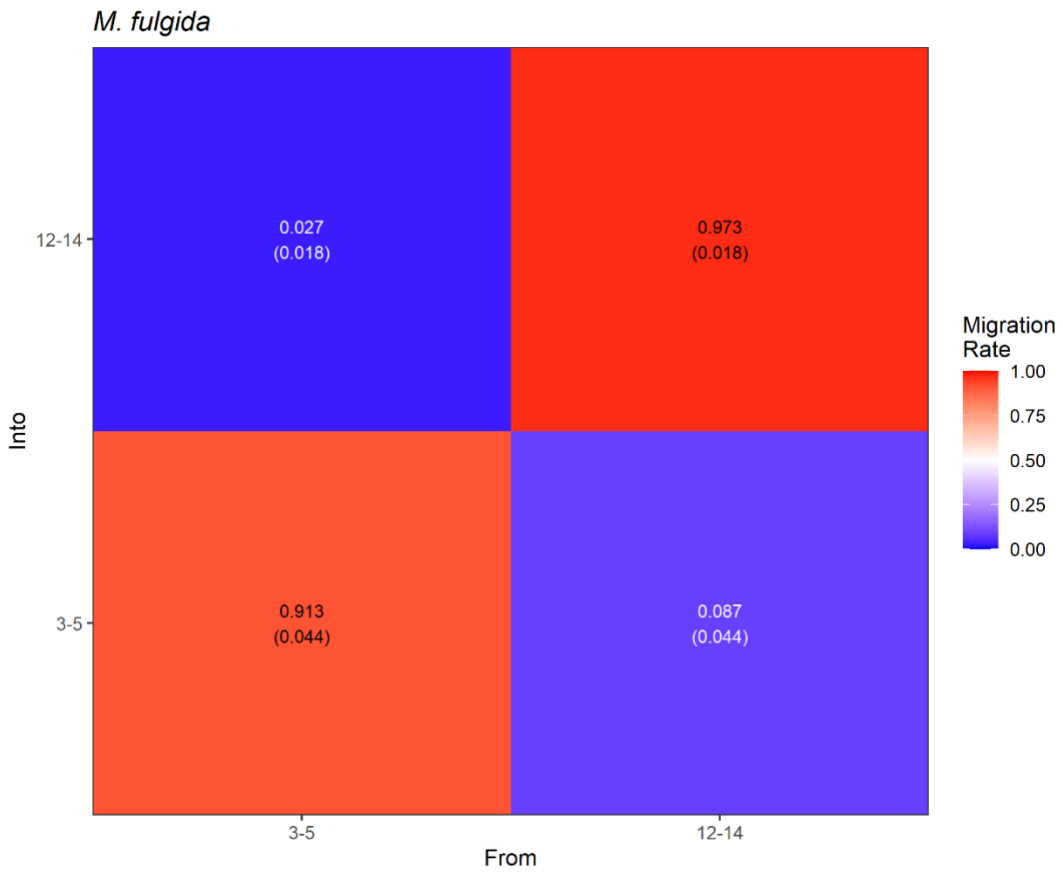

Figure S.4. Migration matrix for *M fulgida*. Posterior means (standard errors), estimated using Bayesian assignment, represent the fraction of individuals at locations along the y-axis that are migrants derived from locations along the x-axis per generation. See Figure S.1 for sample locations corresponding the labels along the x- and y-axes.

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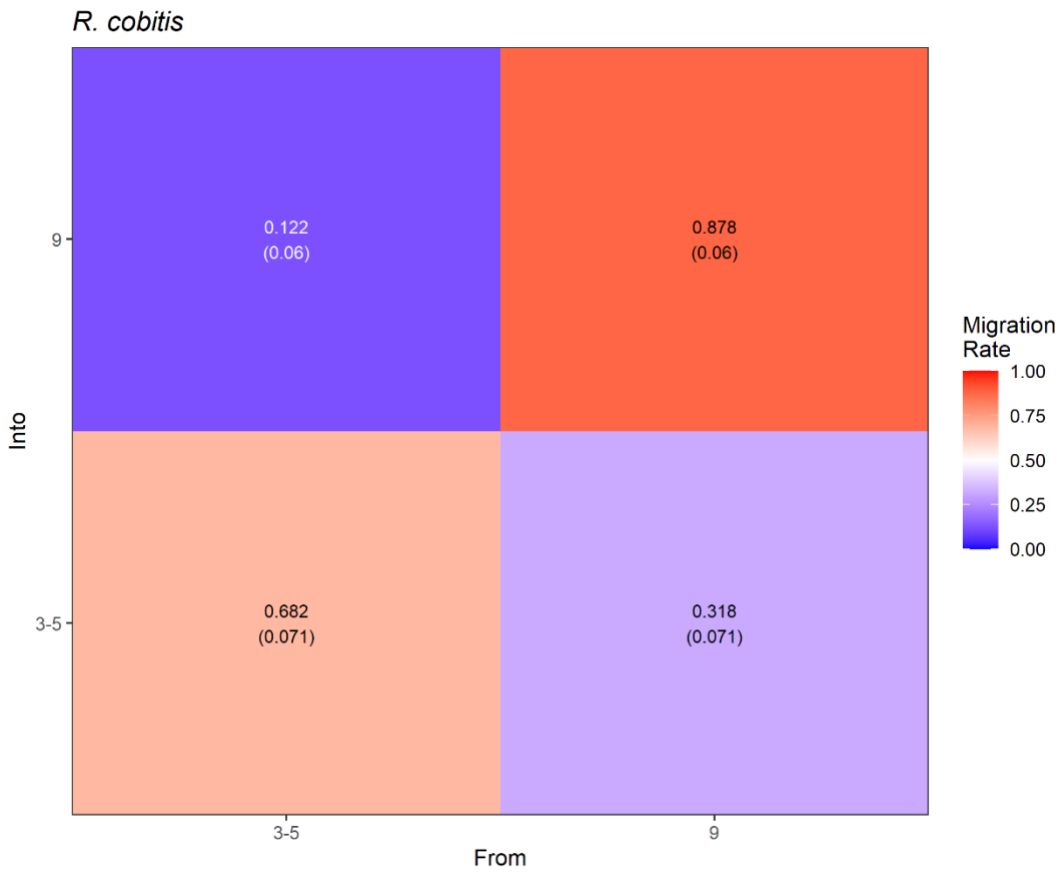

Figure S.5. Migration matrix for *R. cobitis*. Posterior means (standard errors), estimated using Bayesian assignment, represent the fraction of individuals at locations along the y-axis that are migrants derived from locations along the x-axis per generation. See Figure S.1 for sample locations corresponding the labels along the x- and y-axes. Note, 2010 genotypes were augmented with genotypes sampled in 2009 (see Pilger et al. 2015) to increase the number of runs that reached convergence.

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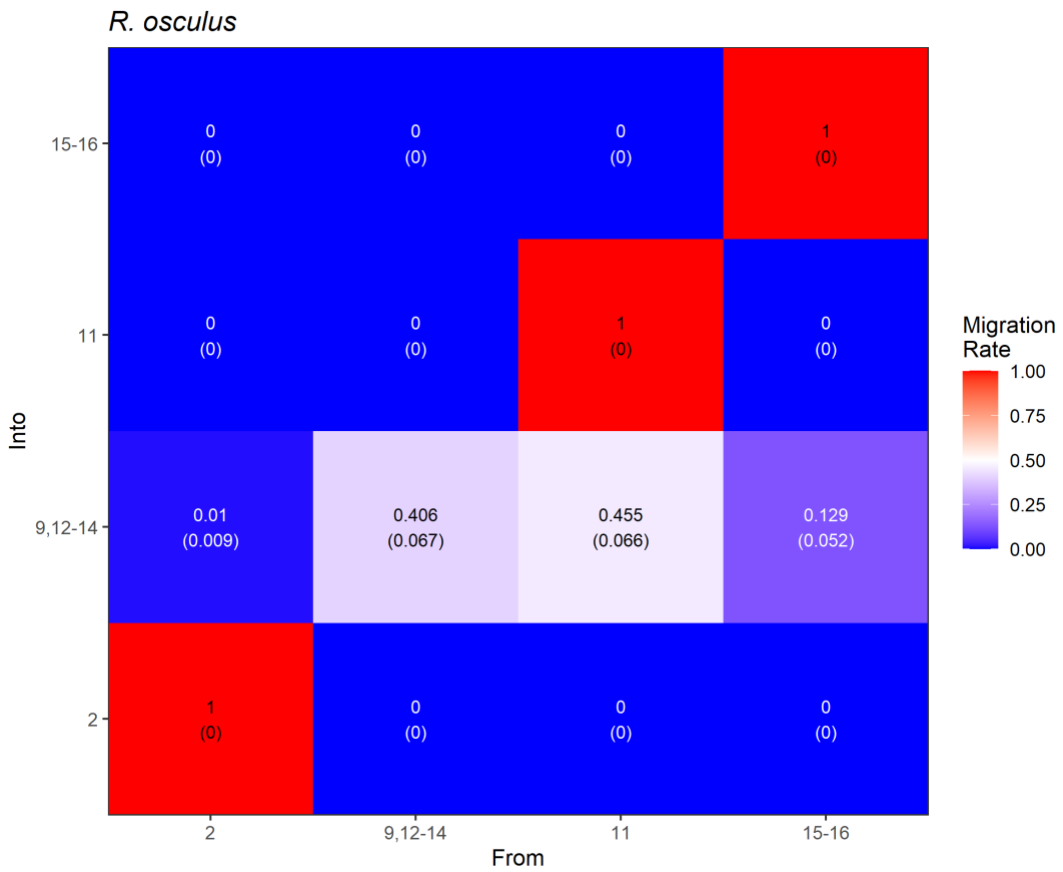

Figure S.6. Migration matrix for *R. osculus*. Posterior means (standard errors), estimated using Bayesian assignment, represent the fraction of individuals at locations along the y-axis that are migrants derived from locations along the x-axis per generation. See Figure S.1 for sample locations corresponding the labels along the x- and y-axes.

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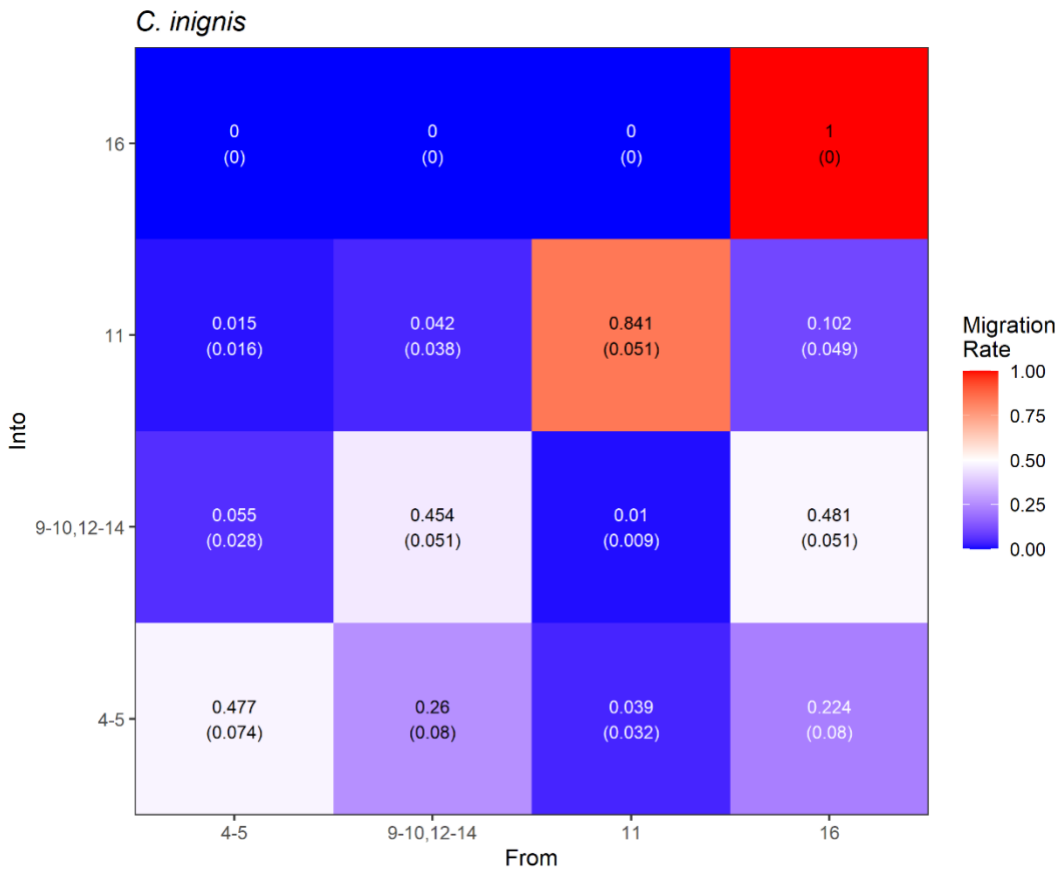

Figure S.7. Migration matrix for *C. insignis*. Posterior means (standard errors), estimated using Bayesian assignment, represent the fraction of individuals at locations along the y-axis that are migrants derived from locations along the x-axis per generation. See Figure S.1 for sample locations corresponding the labels along the x- and y-axes.

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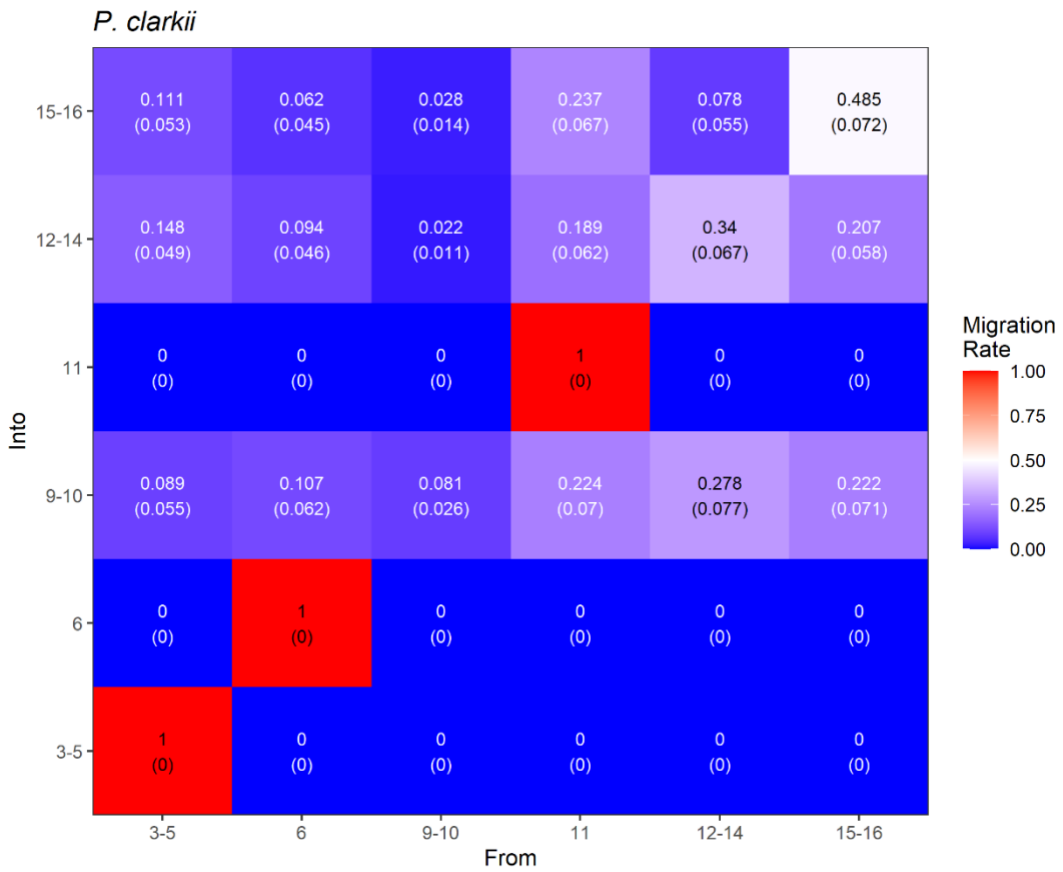

Figure S.8. Migration matrix for *P. clarkii*. Posterior means (standard errors), estimated using Bayesian assignment, represent the fraction of individuals at locations along the y-axis that are migrants derived from locations along the x-axis per generation. See Figure S.1 for sample locations corresponding the labels along the x- and y-axes.

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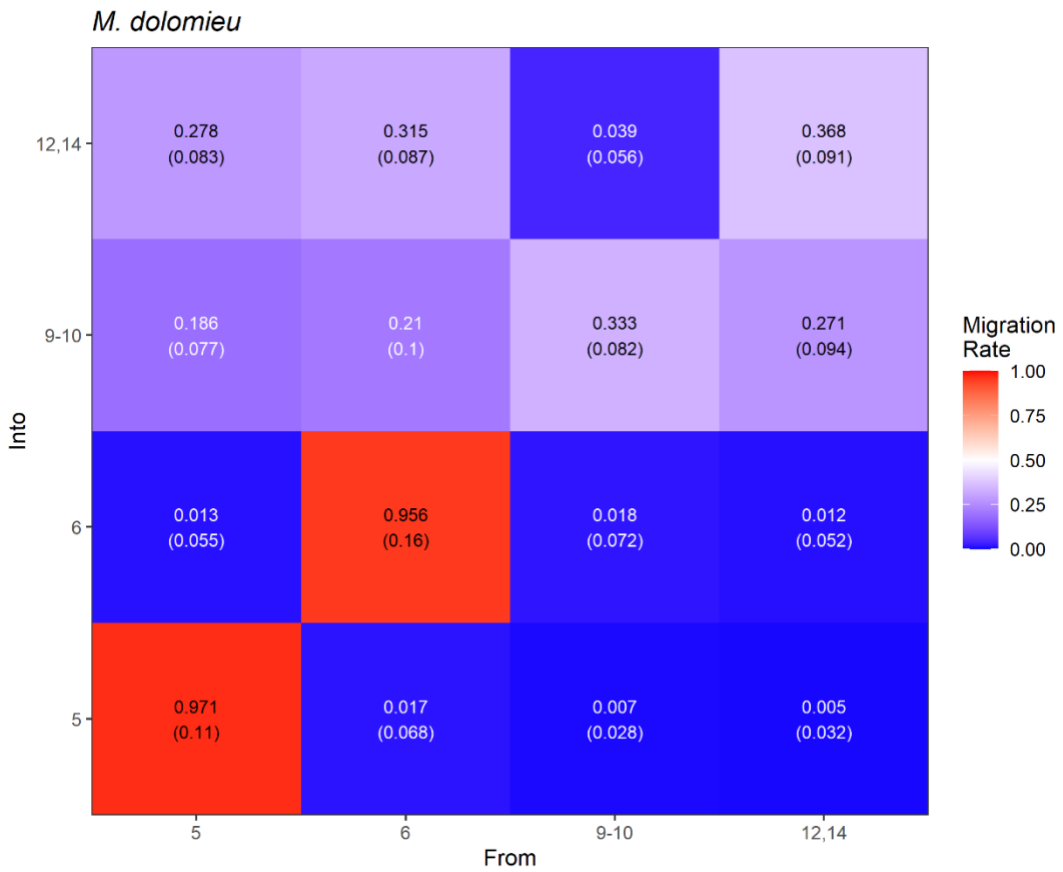

Figure S.9. Migration matrix for *M. dolomieu*. Posterior means (standard errors), estimated using Bayesian assignment, represent the fraction of individuals at locations along the y-axis that are migrants derived from locations along the x-axis per generation. See Figure S.1 for sample locations corresponding the labels along the x- and y-axes.

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Estimates of effective number of breeders (N_b)

Table S.4. Effective number of breeders for Gila River fishes

Species	SampleCode	SiteNumber	Sample.n	Sibship Nb	Lower 95% CI	Upper 95% CI	LD Nb	Lower 95% CI	Upper 95% CI
Agosia chrysogaster	WF	13	36	35	22	60	84.7	43.7	447.1
	MF	14	30	54	31	101	254.4	254.4	inf
	HB	12	30	53	32	93	172.3	172.3	inf
	BCA	15	30	40	75	21	101.2	101.2	inf
	GV	9	28	50	30	89	128.5	128.5	inf
	SAP	8	30	53	33	95	162.2	67.8	inf
	TURC	7	30	58	37	99	69.9	42.3	165
	GF	6	31	58	37	99	141.7	141.7	inf
	RS	5	30	56	35	103	126.4	126.4	inf
	BA	4	32	68	43	120	326.1	326.1	inf
	MB	3	29	54	33	92	174.2	74.2	inf
	BLUC	2	30	45	26	75	134	134	inf
	SSC	1	31	62	39	107	121.5	121.5	inf
Gila robusta	WF	13	26	43	25	77	52.8	27.9	210
	MF	14	20	25	14	50	61.5	27.1	inf
	HB	12	3	-	-	-	-	-	-

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Species	SampleCode	SiteNumber	Sample.n	Sibship Nb	Lower 95% CI	Upper 95% CI	LD Nb	Lower 95% CI	Upper 95% CI
Gila robusta	UEF	16	19	22	12	47	58.4	28.5	518.2
	GV	9	12	14	7	39	8.2	5.9	11.4
Meda fulgida	WF	13	30	56	34	96	176.9	176.9	inf
	MF	14	34	43	26	72	82.3	82.3	inf
	HB	12	30	40	24	69	100.2	58.8	279.9
	RS	5	28	69	40	126	90.2	90.2	inf
	BA	4	31	81	52	142	178.7	178.7	inf
	MB	3	17	109	52	4984	50.3	50.3	inf
Rhinichthys cobitis	GV	9	30	67	39	142	140.6	70.7	1352.7
	RS	5	29	74	47	137	249.4	57.9	inf
	BA	4	34	77	50	131	149.6	149.6	inf
	MB	3	19	98	50	788	349.3	349.3	inf
Rhinichthys osculus	UEF	16	19	49	27	120	115.6	115.6	inf
	BCA	15	29	52	33	89	161.1	63.3	inf
	WF	13	26	39	24	68	208.7	68.3	inf
	MF	14	30	45	26	79	117.2	51.5	inf
	HB	12	30	62	39	111	150.6	60.1	inf
	LC	11	32	52	32	94	124.6	65.1	654.4
	GV	9	30	58	35	100	314.7	314.7	inf

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Species	SampleCode	SiteNumber	Sample.n	Sibship Nb	Lower 95% CI	Upper 95% CI	LD Nb	Lower 95% CI	Upper 95% CI
Rhinichthys osculus	BLUC	2	30	39	24	68	88.7	88.7	inf
Catostomus insignis	UEF	16	33	49	30	83	168.8	168.8	inf
	BCA	15	9	-	-	-	-	-	-
	WF	13	30	40	25	71	116.7	59.4	865.9
	MF	14	31	49	31	86	209.3	209.3	inf
	HB	12	30	47	28	85	297.1	297.1	inf
	LC	11	31	9	4	24	9.5	7.6	11.7
	LEF	10	30	50	31	88	237.5	79.8	inf
	GV	9	28	42	24	73	193.1	81	inf
	GF	6	7	-	-	-	-	-	-
	RS	5	28	44	27	78	724.8	94.9	inf
	BA	4	31	58	36	101	343.1	78.4	inf
	MB	3	3	-	-	-	-	-	-
Pantosteus clarkii	WF	13	30	62	40	109	268.7	95.4	inf
	MF	14	32	79	49	135	299.8	70.1	inf
	HB	12	31	55	34	98	88.9	53.5	217.6
	LC	11	31	25	14	46	39.6	29.7	56.3
	UEF	16	32	47	30	77	227.2	90.5	inf
	BCA	15	35	38	23	65	50.3	34.4	84.1

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Species	SampleCode	SiteNumber	Sample.n	Sibship Nb	Lower 95% CI	Upper 95% CI	LD Nb	Lower 95% CI	Upper 95% CI
	LEF	10	29	42	25	74	65.1	42.8	122.9
	GV	9	29	51	32	94	34.6	22.5	63.2
	GF	6	14	23	12	56	39.3	21.8	126
	RS	5	30	51	32	85	100.4	55.1	376
	BA	4	31	39	24	69	136.4	62.4	inf
	MB	3	20	51	29	106	422.8	72.7	inf
Ameiurus natalis	UEF	16	32	13	7	30	36.3	2.5	inf
	MF	14	37	15	8	31	16.8	3.7	127.1
	HB	12	12	11	5	34	100.4	2.1	inf
	LEF	10	16	11	5	28	42.8	5.3	inf
	GV	9	21	16	8	35	36.5	4.5	inf
	BA	4	4	-	-	-	-	-	-
Micropterus dolomieu	UEF	16	5	-	-	-	-	-	-
	WF	13	2	-	-	-	-	-	-
	MF	14	35	16	9	34	31.8	13.2	196.8
	HB	12	26	16	8	34	62.9	16.7	inf
	LEF	10	30	14	8	30	12.3	4.6	37.5
	GV	9	29	16	9	33	155.9	18.2	inf
	TURC	7	2	-	-	-	-	-	-

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Species	SampleCode	SiteNumber	Sample.n	Sibship Nb	Lower 95% CI	Upper 95% CI	LD Nb	Lower 95% CI	Upper 95% CI
Micropterus dolomieu	GF	6	30	18	10	38	21.1	8.4	122.4
	RS	5	31	17	10	38	23.8	6.7	inf
	BA	4	5	-	-	-	-	-	-

Note-Effective number of breeders (N_b) was estimated using the sibship method (Wang 2009) implemented in COLONY (Jones and Wang 2010) and the linkage disequilibrium method (LD, Hill 1981) implemented in LDNE (Waples and Do 2008). A dash (-) indicates the sample size was not sufficient to estimate N_b. Site numbers correspond to Figure S.1.

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References

Faubet, P. and O. E. Gaggiotti. 2008. A new Bayesian method to identify the environmental factors that influence recent migration. Genetics 178:1491-1504.

Hill, W. G. 1981. Estimation of effective population size from data on linkage disequilibrium. Genetics Research 38:209-216.

Hines, J. E. (2006). PRESENCE - Software to estimate patch occupancy and related parameters. SGS-PWRC. http://www.mbr-pwrc.usgs.gov/software/presence.html.

Jones, O. R., and J. Wang. 2010. COLONY: a program for parentage and sibship inference from multilocus genotype data. Molecular Ecology Resources 10:551-555.

Olden, J. D., N. L. Poff, and K. R. Bestgen. 2006. Life-history strategies predict fish invasions and extirpations in the Colorado River Basin. Ecological Monographs 76:25-40.

Pilger, T. J., K. B. Gido, D. L. Propst, J. E. Whitney, and T. F. Turner. 2015. Comparative conservation genetics of protected endemic fishes in an arid-land riverscape. Conservation Genetics 16:875-888.

Pilger, T. J., K. B. Gido, D. L. Propst, J. E. Whitney, and T. F. Turner. 2017. River network architecture, genetic effective size and distributional patterns predict differences in genetic structure across species in a dryland stream fish community. Molecular Ecology 26:2687-2697.

Wang, J. 2009. A new method for estimating effective population sizes from a single sample of multilocus genotypes. Molecular Ecology 18:2148-2164.

Waples, R. S., and C. Do. 2008. LDNE: a program for estimating effective population size from data on linkage disequilibrium. Molecular Ecology Resources 8:753-756.

Whitney, J. E., K. B. Gido, T. J. Pilger, D. L. Propst, and T. F. Turner. 2016. Metapopulation analysis indicates native and non-native fishes respond differently to effects of wildfire on desert streams. Ecology of Freshwater Fish 25:376-392.